

Define Problem Statements

Phase: Brainstorming & Ideation

Project: ShopEEZ | Date: June 2026 | Stack: MongoDB · Express · React · Node.js

1. Customer Problem Statements

Problem 1: Poor Mobile Shopping Experience

Context: Most consumers browse products on mobile devices.

Problem: Generic e-commerce UIs use cluttered desktop-first layouts that break on smaller screens, leading to high cart abandonment rates.

ShopEEZ Solution: Fully responsive Vanilla CSS Flexbox/Grid layout — adapts cleanly from 320px mobile to 1440px+ desktop viewports.

Problem 2: Lost Shopping Cart on Browser Refresh

Context: Users add items to their cart, leave, or refresh the page.

Problem: Cart state is lost on session expiry or page reload when stored only in memory.

ShopEEZ Solution: Redux Toolkit cartSlice persists the entire cart to localStorage automatically on every update.

Problem 3: Lack of Product Discoverability

Context: Stores with 50+ products need powerful browsing tools.

Problem: No filtering or search means users scroll endlessly through irrelevant products.

ShopEEZ Solution: Full-text search + live-filtering by category, price range, rating, and sorting via GET /api/products query params.

2. Administrator Problem Statements

Problem 4: Complex or Decoupled Inventory Management

Context: Store admins need to update prices, stock levels, and product images regularly.

Problem: Most admin panels are either overly complex or disconnected from the live storefront.

ShopEEZ Solution: Integrated Admin Dashboard at /admin/dashboard with direct CRUD over the MongoDB products collection.

Problem 5: No Visibility into Order Fulfillment

Context: Admins need to track which orders are being processed, shipped, or delivered.

Problem: Without a centralized order management view, fulfillment is manual and error-prone.

ShopEEZ Solution: /admin/orders page with status toggle: processing !' shipped !' delivered !' cancelled, updating MongoDB directly.